DevOps Assignment Summary

Simple explanations of what I did in each set of assignments

Git

I used Git to keep track of changes to my files and uploaded them to GitHub. I practised working on separate branches, saving unfinished work, and combining changes.

Terraform

I used code to create servers and networks on AWS instead of setting everything up manually. I also gave a server a fixed IP address and automatically installed Apache.

Ansible

I used Ansible to install software such as Java, MySQL, and web servers on several machines automatically. I also updated website files and separated machines into test and production groups.

Docker

I ran an Apache web server inside a container - a separate environment for an application. I saved it as a reusable image, uploaded it to Docker Hub, and used it on another machine.

Jenkins

I set up jobs that ran automatically when I pushed code changes. They copied files to a test server first, then to the production server if the test job succeeded.

Kubernetes

I set up three machines working together and ran several copies of NGINX, a web server. I increased the copies from three to five and configured how users could reach them.